Junbo Zhao

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Personal Website: https://a-lincui.github.io

EDUCATION

Shanghai Jiao Tong University, Shanghai, China 09/2025-now

Ph.D. Student in Computer Science and Technology

Tsinghua University, Beijing, China 09/2021-06/2024

Master's degree in Electronic and Information Engineering

Thesis Title: Automatic Design Method of Efficient Neural Network Based on Performance Prediction and Weight sharing

Tsinghua University, Beijing, China 09/2017-06/2021

Bachelor's degree in Engineering Physics

PUBLICATION

- **TA-GATES: An Encoding Scheme for Neural Network Architectures.** Xuefei Ning*, Zixuan Zhou*, **Junbo Zhao**, Tianchen Zhao, Yiping Deng, Changcheng Tang, Shuang Liang, Huazhong Yang, Yu Wang. **NeurIPS2022**.
- **Dynamic Ensemble of Low-fidelity Experts: Mitigating NAS "Cold-Start". Junbo Zhao***, Xuefei Ning*, Enshu Liu, Binxin Ru, Zixuan Zhou, Tianchen Zhao, Chen Chen, Jiajin Zhang, Qingmin Liao, Yu Wang. **AAAI2023**.
- **Ensemble-based Reliability Enhancement for Edge-Deployed CNNs in Few-shot Scenarios.** Zhen Gao, Shuang Liu, **Junbo Zhao**, Xiaofei Wang, Yu Wang, Zhu Han. **TMLCN2024**.
- **TB-STC: Transposable Block-wise N:M Structured Sparse Tensor Core.** Jun Liu, Shulin Zeng, **Junbo Zhao**, Li Ding, Zeyu Wang, Jinhao Li, Zhenhua Zhu, Xuefei Ning, Chen Zhang, Yu Wang, Guohao Dai. **HPCA2025**.
- **Discovering Robust Convolutional Architecture at Targeted Capacity: A Multi-Shot Approach.** Xuefei Ning*, **Junbo Zhao***, Wenshuo Li, Tianchen Zhao, Yin Zheng, Huazhong Yang, Yu Wang. arXiv:2012.11835.

PROFESSIONAL EXPERIENCES

Infinigence Technology Co., Ltd. 08/2023-12/2023

Algorithm Researcher

- **Model Efficiency Evaluation**: Evaluated the efficiency of large language models (e.g., GLM-6B, OPT-6.7B) on domestically produced chips (e.g., Tianyue, Haifeike) in China, providing crucial data for lightweight large language model design.
- **Chip Testing Solution**: Contributed to the design and development of a unified chip testing solution.

Novauto Technology Co., Ltd. 03/2023-07/2023

Algorithm Researcher

- **Hardware-Aware NAS Toolchain Development**: Developed a hardware-aware toolchain for neural network architecture optimization, enhancing computational efficiency and performance.
- **Toolchain Validation**: Validated the toolchain's effectiveness through practical applications in visual tasks (e.g., image denoising, object detection), analyzing its advantages and limitations for further improvement.

LANGUAGE& SKILLS

- IELTS: Overall: 7.5 (Reading: 9.0, Listening: 8.0, Speaking: 5.5, Writing: 6.5)
- Software: Microsoft Office, Git, Vim, Linux, Python, C/C++